UNIVERSITY OF AMSTERDAM

REPORT

Infectious disease modelling with SIR models

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1 Introduction

Over the past 100 years epidemics such as the Spanish flu, HIV/AIDS and cholera have affected the health and survival of the population worldwide (Bloom & Cadarette, 2019). Most recently, the COVID-19 epidemic resulted in 776.8 million reported cases and 7 million reported deaths by November 2024 (World Health Organization, 2024). An understanding of the mechanisms underlying infectious disease spread an persistence is needed for development of epidemiological intervention strategies.

Mathematical compartmental models allow studying infectious disease spread and persistence by sorting individuals in different health states and describing how these compartments change over time (Keeling & Rohani, 2011, Chapter 1). Using these models effects of intervention strategies such as vaccination, quarantine and culling on disease transmission at population level can be studied (Keeling & Rohani, 2011, Chapter 1). In 1927 Kermack and McKendrick developed the mathematical compartmental SIR-model in which Susceptible (S), Infected (I) and Recovered (R) individuals are described in a closed population by a system of differential equations as a function over time. This model has been useful for establishing if infection spread is growing by calculating the basic reproductive ratio R_0, this is the average number of individuals that one infected individual would infect in a susceptible population (Keeling & Rohani, 2011, Chapter 2). If $R_0 > 1$, an infection can invade (Keeling & Rohani, 2011, Chapter 2). Although its simple structure makes the SIR model useful for studying the main dynamics of an epidemic, this simplicity also limits how closely it represents real populations (Keeling & Rohani, 2008, Chapter 2). The SIR compartmental model homogenises a population and leaves out individual differences to infectious diseases such as age: older people are more susceptible to infection, have longer recovery time and generally have less social contacts than younger people. (Keeling & Rohani, 2008, Chapter 2). Introducing additional biological differences can make a model more representative of a real epidemic, but this also increases its complexity and the amount of data required to describe the system (Keeling & Rohani, 2008, Chapter 1). Thus, more detailed models may be harder to interpret, while simpler models can provide clearer insight with less accuracy (Keeling & Rohani, 2008, Chapter 1). This conflict of interest becomes important when models are used to predict epidemic spread. A systematic review of COVID-19 modelling studies found a difference between predicted and observed epidemic outcomes (Gnanvi et al., 2021). Showing that when interpreting epidemiologic prediction model assumptions, parameter estimates and the available data should be considered. This report therefore investigates how changes in epidemiological parameters, control strategies, population dynamics and seasonal transmission affect the behaviour of SIR-based epidemic models.

First, the basic SIR model is numerically integrated to investigate the epidemic threshold and its phase space behaviour. Then transmission and recovery parameters are fitted to influenza outbreak data. The model is then extended to investigate vaccination strategies, demography and infection induced mortality. Lastly, an alternative compartmental model and the influence of seasonal variation on infection are researched. To conclude, these analyses demonstrate how progressively modifying the assumptions of a SIR model changes its predicted dynamics.

2 Theory

2.1 Basic SIR model

2.1.1 Model description and assumptions

The basic SIR model that was described by Kermack and McKendrick in 1927 divides the population in three compartments that describe different health states in the context of an acute infectious disease: 1) S – susceptible. This compartment contains the fraction of the population that is currently susceptible to the infection. 2) I – Infectious. This compartment contains the

fraction of the population that is currently infectious. 3) R – Recovered. This compartment contains the fraction of the population that is recovered from the infection (Keeling & Rohani, 2008, Chapter 2). The movement from one compartment to the other is visualised in Flow diagram 1.

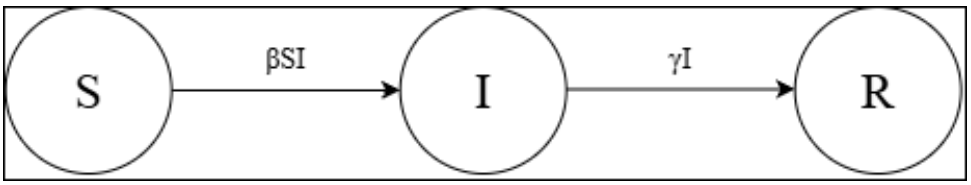
This basic SIR model has the following assumptions:

1) The population size is constant (Keeling & Rohani, 2008, Chapter 2). This makes it possible to study population fractions in each compartment.

2) There are no births, deaths and migration in this population, this is also known as a closed population (Keeling & Rohani, 2008, Chapter 2).

3) There is no pre-existing immunity, thus $R0 = 0$ (Keeling & Rohani, 2008, Chapter 2)

4) There is homogeneous mixing, every individual has equal contact (Keeling & Rohani, 2008, Chapter 2)



Flow diagram 1. Flow diagram displays basic SIR model. Arrows display direction of movement from one compartment to the other. Above the arrows is shown the transition rates of infection and recovery (Keeling & Rohani, 2008, Chapter 2).

2.1.2 Model parameters and equations

When infection dynamics are studied with absolute numbers of individuals, X, Y and Z can be used to represent the numbers of susceptible, infected and recovered individuals (Keeling & Rohani, 2008, Chapter 2). Two parameters determine the move from one compartment to the other. The first being β, which is the transmission parameter that combines the rate of potentially infectious contacts and the probability that transmission occurs during a contact (Keeling & Rohani, 2008, Chapter 2). Second being γ, which is the recovery rate, while $1/\gamma$ represents the average infectious period (Keeling & Rohani, 2008, Chapter 2). The following SIR differential equations follow the set assumptions and use these parameters:

$$\frac{dS}{dt} = -\beta SI,$$
$$\frac{dI}{dt} = \beta SI - \gamma I,$$
$$\frac{dR}{dt} = \gamma I.$$

(Kermack and McKendrick, 1927. Keeling & Rohani, 2008, Chapter 2.)

Because S, I and R are fractions of a constant population,

$S(t) + I(t) + R(t) = 1$

(Keeling & Rohani, 2008, Chapter 2). The force of infection λ, describes the rate at which susceptible individuals acquire infection. For the frequency dependent SIR formulation:

$$\lambda = \beta I,$$

or equivalently $\lambda = \frac{\beta Y}{N}$ absolute population numbers are used (Keeling & Rohani, 2008, Chapter 2). Since SIR modelling start with $R0 = 0$:

$$S(0) + I(0) = 1$$

2.1.3 Epidemic threshold and basic reproductive number

The threshold phenomenon and following the reproductive ratio are factors that determine if infection will invade (Keeling & Rohani, 2008, Chapter 2). Kermack and McKendrick (1927) found the threshold phenomenon from the infected equation. They rewrote $\frac{dI}{dt} = \beta SI - \gamma I$ to:

$$\begin{aligned} \frac{dI}{dt} &= I(\beta S - \gamma) \\ \frac{dI}{dt} &= \beta I\left(S - \frac{\gamma}{\beta}\right) \end{aligned}$$

From this it can be concluded that the initial $\frac{dI}{dt}$ will only increase if

$$S(0) > \frac{\gamma}{\beta}$$

(Kermack and McKendrick, 1927. Keeling & Rohani, 2008, Chapter 2.). The relative removal rate, $\frac{\gamma}{\beta}$, needs to be small enough for the infected fraction to grow (Keeling & Rohani, 2008, Chapter 2). The inverse is the basic reproduction number

$$R_0 = \frac{\beta}{\gamma}.$$

Therefore, the condition for initial infection growth is

$$R_0 S(0) > 1$$

When the starting condition $S(0) \approx 1$, $R_0 > 1$ can be used as condition (Keeling & Rohani, 2008, Chapter 2).

2.1.4 Initial conditions and phase space analysis

A phase space plot represents the state of a dynamical system by plotting one variable against another instead of making to plots against time. A solution of the differential equations therefore forms a trajectory through the phase plane (MIT OpenCourseWare, 2011). In this report, the susceptible fraction $S$is plotted on the x-axis and the infected fraction I on the y-axis. Each point on a trajectory consequently represents the combination $(S(t), I(t))$. The trajectory of S and I can be obtained directly from the SIR equations by eliminating t:

$$\frac{dI}{dS} = \frac{\frac{dI}{dt}}{\frac{dS}{dt}} = \frac{\beta SI - \gamma I}{-\beta SI} = -1 + \frac{\gamma}{\beta S} = -1 + \frac{1}{R_0 S}.$$

Different initial values $S(0)$ and $I(0)$ generate different trajectories. The infected fraction reaches its maximum when $\frac{dI}{dt} = 0$. This occurs at:

$$S = \frac{\gamma}{\beta} = \frac{1}{R_0}$$

2.1.5 Parameter estimation from outbreak data

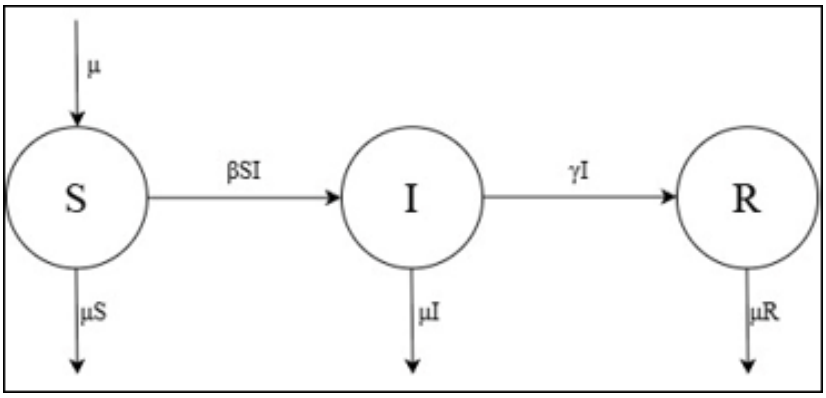
From observed epidemic data parameter estimation for β and γ can be done with the least squares method (Keeling & Rohani, 2008, Chapter 2). This method finds parameter values by minimizing the squared differences between observed and modelled values (van de Geer, 2005). Consecutively R_0 can be calculated from the estimated β and γ.

2.1.6 Vaccination in the basic SIR model

With a vaccination plan, the vaccinated fraction of the susceptible individuals can be transferred into the recovered compartment. This model assumes that immunity achieved through vaccination or disease is biologically identical (Keeling & Rohani, 2008, Chapter 8).

2.2 SIR model with demography

To work out the basic SIR model further, demography can be added to the model. Demography is a birth and mortality rate (Keeling & Rohani, 2008, Chapter 2). Whilst basic SIR models can be suitable for modelling short lasting diseases, in which birth and mortality do not play a role, SIR modelling with demography can give more insight on how infection dynamics persist in diseases where new susceptible are added by birth and people dying from each category (Keeling & Rohani, 2008, Chapter 2). The Flow diagram 2 visualises the SIR model with demography.



Flow diagram 2. Flow diagram displays SIR model with demography. Arrows display direction of movement from one compartment to the other. Above the arrows is shown the transition rates of infection and recovery. The arrow labelled μ represents births into the susceptible compartment, while μS, μI, and μR represent natural deaths from the susceptible, infected and recovered compartments (Keeling & Rohani, 2008, Chapter 2).

Parameter μ represents the birth rate (Keeling & Rohani, 2008, Chapter 2). Since population remains constant as in the basic SIR, the death rate is equal to μ. Since people can die from each compartment, the death rate is proportional to the population size of the compartment making $\mu S + \mu I + \mu R = \mu$ (Keeling & Rohani, 2008, Chapter 2). Adding demography to the differential equations:

$$\begin{aligned} \frac{dS}{dt} &= \mu - \beta SI - \mu S \\ \frac{dI}{dt} &= \beta SI - \gamma I - \mu I \\ \frac{dR}{dt} &= \gamma I - \mu R \end{aligned}$$

2.2.1 The basic reproductive number in SIR with demography

The R_0 in the SIR in demography differs since people leave the infectious category quicker due to death (Keeling & Rohani, 2008, Chapter 2). The number of secondary infectives is dependent on transmission rate β times the average time an individual spends in the infected compartment (Keeling & Rohani, 2008, Chapter 2). This is dependent on μ and $\frac{1}{\gamma}$, so the average time an individual spends in the infected compartment is $\frac{1}{\mu+\gamma}$ (Keeling & Rohani, 2008, Chapter 2). Leading to R_0:

$$R_0 = \beta \frac{1}{\mu + \gamma} = \frac{\beta}{\mu + \gamma}$$

2.2.2 Disease free and endemic equilibrium

To understand how disease develops over time it is important to know when disease states become stagnant, also known as at equilibrium (Keeling & Rohani, 2008, Chapter 2). At equilibrium, the fractions of susceptible, infected, and recovered individuals remain constant over time (Keeling & Rohani, 2008, Chapter 2). Therefore, all three derivatives are set equal to zero:

$$\frac{dS}{dt} = \frac{dI}{dt} = \frac{dR}{dt} = 0.$$

The equilibrium values of the three compartments are denoted by S^*, I^*, and R^*, where * indicates the value of the compartment at equilibrium (Keeling & Rohani, 2008, Chapter 2). The SIR model with demography has a disease free equilibrium in which the infection is absent and the population is entirely susceptible (Keeling & Rohani, 2008, Chapter 2). This equilibrium is

$$(S^*, I^*, R^*) = (1, 0, 0).$$

An endemic equilibrium is a second possible equilibrium state in which infection remains present in the population, so $I^* > 0$(Keeling & Rohani, 2008, Chapter 2).

To derive the endemic equilibrium, the infected equation is first set to zero, giving $0 = \beta S^* I^* - (\gamma + \mu) I^*$, which can be factored into $I^* (\beta S^* - (\gamma + \mu)) = 0$ (Keeling & Rohani, 2008, Chapter 2). This equation is satisfied when either $I^* = 0$ or $\beta S^* - (\gamma + \mu) = 0$; the first solution represents the disease free equilibrium, so for the endemic equilibrium the second solution is used (Keeling & Rohani, 2008, Chapter 2). Rearranging gives $S^* = \frac{\gamma+\mu}{\beta}$. Because $R_0 = \frac{\beta}{\mu+\gamma}$, this can be written as $S^* = \frac{1}{R_0}$ (Keeling & Rohani, 2008, Chapter 2). Rearranging first gives $\beta S^* I^* = \mu (1 - S^*)$, and then $I^* = \mu \frac{(1-S^*)}{(\beta S^*)}$. Substitution of $S^* = \frac{1}{R_0}$, gives $I^* = \mu \frac{\left(1-\frac{1}{R_0}\right)}{\left(\frac{\beta}{R_0}\right)} = \frac{\mu}{\gamma}\left(1 - \frac{1}{R_0}\right) = \frac{\mu}{\gamma}(R_0 - 1)$

(Keeling & Rohani, 2008, Chapter 2). For this endemic equilibrium I^*cannot be negative, thus $R_0 > 1$. The endemic equilibrium is:

$$(S^*, I^*, R^*) = \left(\frac{1}{R_0}, \frac{\mu}{\gamma}(R_0 - 1), 1 - \frac{1}{R_0} - \frac{\mu}{\beta}(R_0 - 1)\right)$$

(Keeling & Rohani, 2008, Chapter 2).

2.2.3 Damped oscillations and Fourier analysis

When $R_0 > 1$, the SIR dynamics will reach endemic equilibrium (Keeling & Rohani, 2008, Chapter 2). Before reaching the endemic equilibrium damped oscillations occur, where the susceptible and infected fractions move above and below their equilibrium values. (Keeling & Rohani, 2008, Chapter 2). The oscillations are caused by the susceptible and infected compartments reacting to each other. When infection increases, the susceptible fraction decreases, in turn infection decreases. Births also adds susceptible, causing another increase in infection. Because the equilibrium is stable, these repeated changes become smaller over time. The mathematical stability analysis shows that this oscillatory behaviour is related to a pair of complex eigenvalues (Keeling & Rohani, 2008, Chapter 2). The negative real part causes the oscillations to decrease, while the imaginary part determines their period (Keeling & Rohani, 2008, Chapter 2). The approximate period is

$$T \approx 2\pi\sqrt{AG}$$

where $A = \frac{1}{\mu(R_0-1)}$ is the mean age at infection and $G = \frac{1}{\mu+\gamma}$ represents the infectious period (Keeling & Rohani, 2008, Chapter 2).

The oscillations can be studied using Fourier analysis. Fourier analysis can be used to find repeating patterns in a signal by separating it into different frequencies (Vretblad, 2003). The Fourier transform is applied to the infected fraction $I(t)$ to find the frequency of the oscillations.

2.2.4 Infection induced mortality

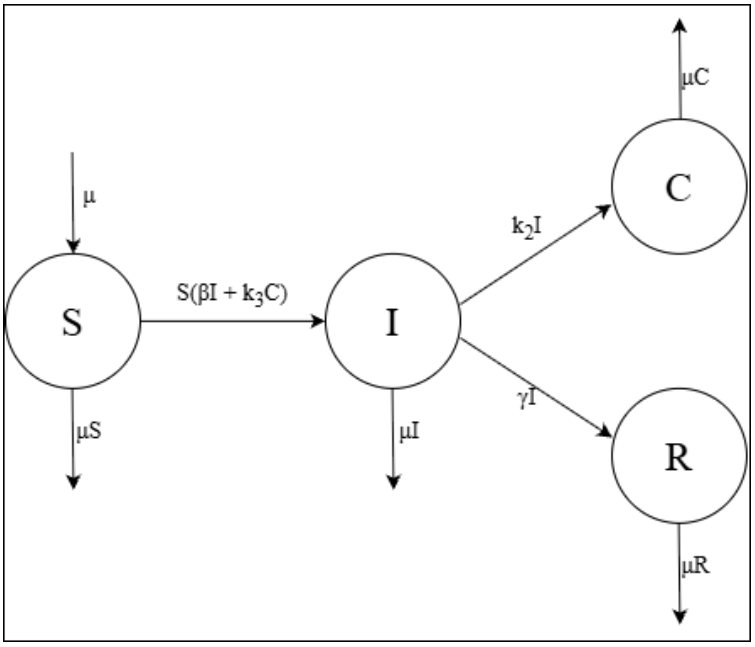
To study diseases that can be fatal, an infection induced mortality factor is introduced. This is ρ, the probability of an infected individual dying from infection (Keeling & Rohani, 2008, Chapter 2). Since infected individuals can leave I through recovery γ, natural death μ or infection induced death m, the mortality probability is $\rho = \frac{m}{(m+\gamma+\mu)}$. This gives the infection induced mortality rate

$$m = \frac{\rho}{1-\rho}(\gamma + \mu)$$

(Keeling & Rohani, 2008, Chapter 2). Because infection induced mortality allows the total population size to change the fixed birth rate ν is included, this is independent of the current population size (Keeling & Rohani, 2008, Chapter 2).

2.3 The SICR model

To model diseases that can transition from infected to either a recovered state or chronic state a SICR model can be used (Keeling & Rohani, 2008, Chapter 2). C is the fraction in the chronic compartment, when an individual becomes chronic it stays in this compartment until death. A real life example of a disease that follows this pattern is HIV/AIDS. The Flow diagram 3 visualises the SICR model.



Flow diagram 3. Flow diagram displays SICR model. Arrows display direction of movement from one compartment to the other. Above the arrows is shown the transition rates of infection, recovery and chronic state. The arrow labelled μ represents births into the susceptible compartment, while μS, μI, μR and μC represent natural deaths from the susceptible, infected, recovered and chronic compartments (Keeling & Rohani, 2008, Chapter 2).

This model introduces two new transmission factors k_2 and k_3. k_2 is the rate at which infected individuals become chronic, k_3 is the rate at which chronic individuals infect susceptible individuals. This leads to the following set of differential equations:

$$\begin{aligned}
\frac{dS}{dt} &= \mu - \beta SI - k_3 SC - \mu S \\
\frac{dI}{dt} &= \beta SI + k_3 SC - (\gamma + k_2 + \mu)\, I \\
\frac{dC}{dt} &= k_2 I - \mu C \\
\frac{dR}{dt} &= \gamma I - \mu R
\end{aligned}$$

2.3.1 Seasonality

The transmission rate of an infectious disease β, can be affected by seasonal changes. Transmission increases and decreases during the year, which influences infection dynamics (Keeling & Rohani, 2008, Chapter 5). This effect can be included by making the transmission rate depend on time, using a sinusoidal function (Keeling & Rohani, 2008, Chapter 5):

$$\beta(t) = \beta_0 \left(1 + \beta_1 \cos(\omega t)\right)$$

Here, β_0 is the average transmission rate, β_1 controls how strongly transmission changes over the seasonal cycle and ω controls how quickly the cycle repeats (Keeling & Rohani, 2008, Chapter 5). In this study, seasonality is added to the acute transmission parameter β, parameter k_3 is kept constant as change of season does not affect chronic transmission. Nevertheless, seasonality will give the chronic and recovered fraction oscillations as they are dependent on the infected fraction.

3 Numerical models

Python version 3.14.6 was used for the numerical implementation.

3.1 Numerical integration of ODEs

Numerical integration of the ODEs of the SIR, SIR model with demography and SICR model were done using the odeint function from the scipy.integrate package. The numpy package was used to create timepoints. The matplotlib.pyplot package was used for plotting the ODE solutions.

3.2 Parameter fitting

The parameters β and γ were fitted to observed influenza data from a English boys school (Table 1) using the curve_fit function from the scipy.optimize package. The observed data were stored using numpy. Matplotlib.pyplot was used to plot the fitted model together with the observed data.

Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Number	1	3	8	28	75	221	291	255	235	190	125	70	28	12	5

Table 1. Table displays number of influenza infections in English boys school.

3.3 Vaccination strategies

Two vaccination strategies were used on the fitted SIR model. The first vaccination strategy simulated the routine immunization approach, in which 49.24% of English boys are vaccinated for influenza before the outbreak. This is simulated by having 49.24% of boys in the Recovered compartment at the start of the outbreak (WHO, 2026). The second vaccination strategy simulated the mass vaccination approach, in which all boys get vaccinated as soon as the first infected boy displays influenza symptoms. For influenza, symptoms appear on day two, so the model was solved up to the vaccination day and restarted with all suseptibles moved to the infected compartment (CDC, 2024). Matplotlib.pyplot was used to visualise the results of the vaccination strategies.

3.4 Fourier analysis

Fourier analysis was performed using the numpy.fft package. The fft function was used to calculate the Fourier transform of $I(t)$, while fftfreq function was used to obtain the corresponding frequencies. The numpy.abs function was used to calculate the amplitude of the Fourier values. The Fourier results were visualised using matplotlib.pyplot.

4 Results and discussion

4.1 Basic SIR model and phase space analysis

The Basic SIR was plotted in the no epidemic state and in the epidemic state. In the no epidemic state $\beta = 0.2$ and $\gamma = 0.3$, giving $R_0 = 0.67$. Figure 1 shows that the infected fraction immediately decreases. For the epidemic state $\beta = 0.3$ and $\gamma = 0.1$, giving $R_0 = 3$. Figure 2 shows that the infected fraction grows before decreasing and reaching the epidemic threshold. The results are in line with the theory, when $R_0 > 1$ the infected fraction will grow (Keeling & Rohani, 2008, Chapter 2).

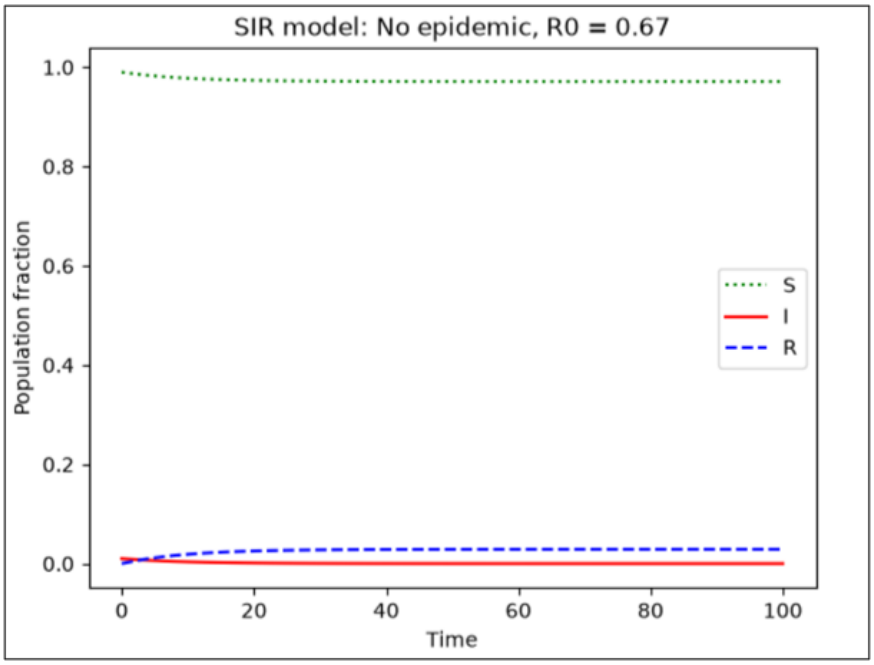


Figure 1. SIR model without epidemic.

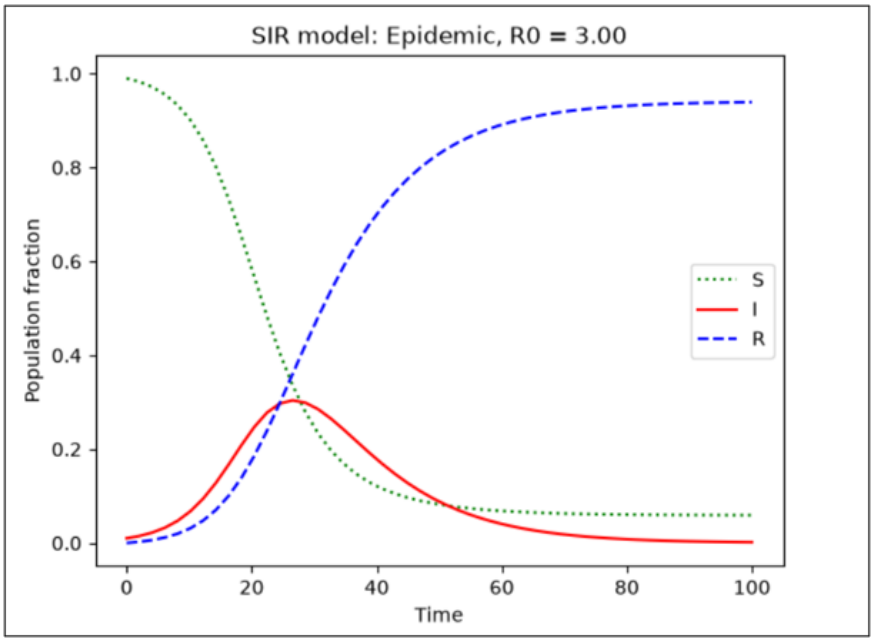


Figure 2. SIR model with epidemic

Figure 3 is the phase space diagram with $R_0 = 3$. The phase space diagram displays the epidemic threshold at $S = \frac{\gamma}{\beta} = \frac{0.1}{0.3} = 0.33$ (Keeling & Rohani, 2008, Chapter 2). The phase space boundary are the maximum values of S and I, on this line $S + I = 1$.

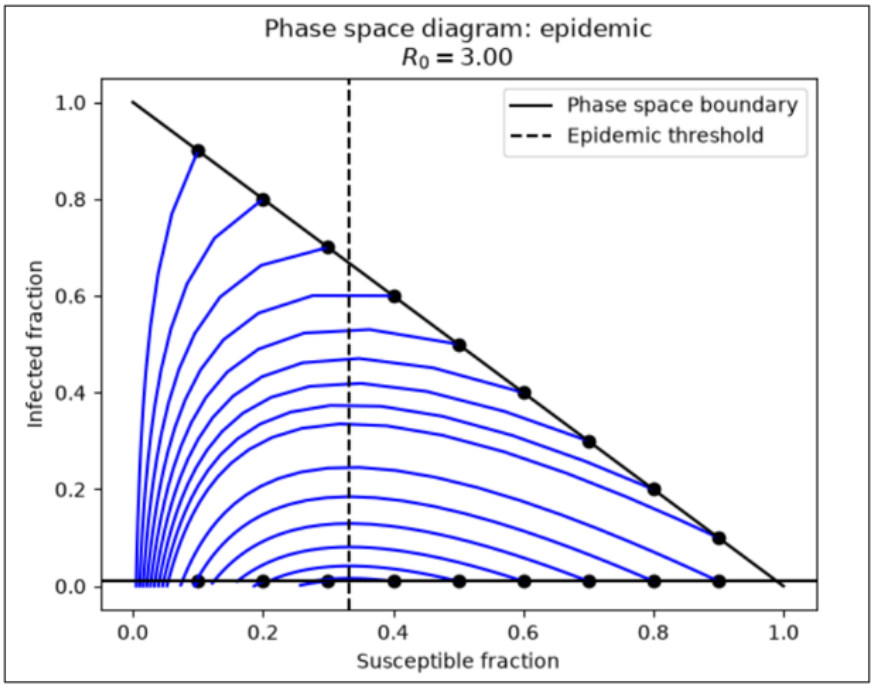


Figure 3. Phase space diagram with $R_0 = 3$. Different S and I starting values are signified with a dot.

4.2 Fitted SIR for influenza data

The SIR model was fitted to the influenza data from an English boys school. This is visualised in **Figure 4**. The fitted transmission rate was $\beta = 1.665$ and the fitted recovery rate was $\gamma = 0.448$. The average infectious period is $\frac{1}{\gamma} = \frac{1}{0.448} = 2.23$ days and $R_0 = \frac{1.665}{0.448} = 3.72$.

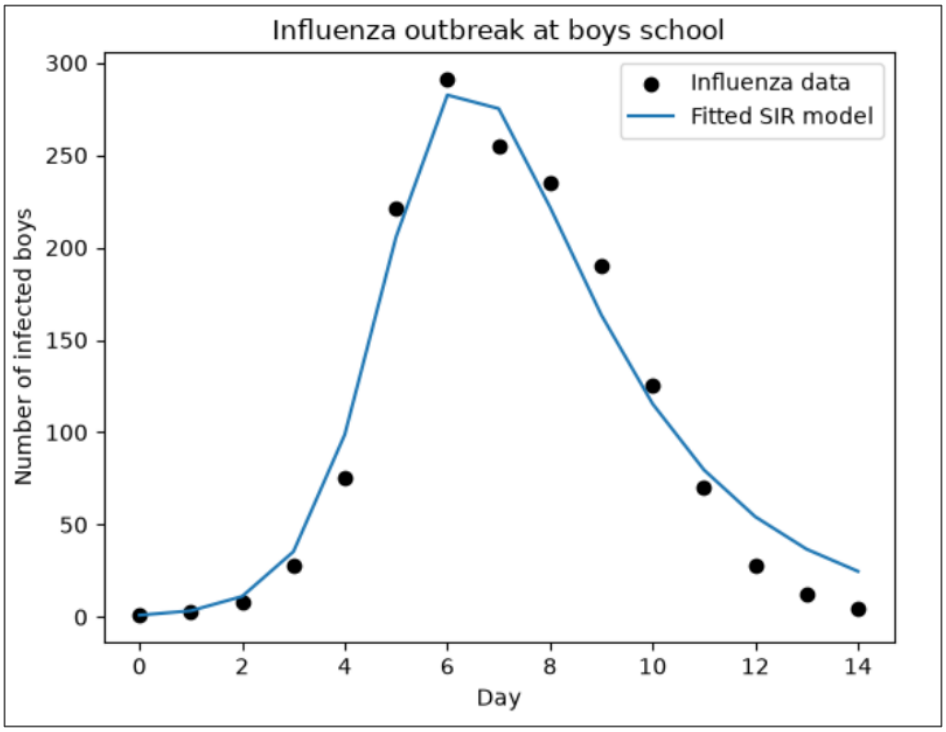


Figure 4. Fitted SIR model to real influenza data from an English boys school.

4.3 Vaccination strategies

The vaccination strategies are visualised in Figure 5. Without vaccination, the fitted model reached a peak around 280 infections. With the routine immunization approach, the peak is around 50 infections. The outbreak also developed slower. The mass vaccination strategy had the strongest effect. After the susceptible population was vaccinated, the infected population went to zero.

These results show that reducing the susceptible population reduces transmission. Infection increases when $S > \frac{\gamma}{\beta}$, vaccination reduces S, making the epidemic die out (Keeling & Rohani, 2008). The mass vaccination strategy was more effective even though it was introduced later, because it moved all remaining susceptible individuals to recovered. This shows that vaccination coverage influences infection growth.

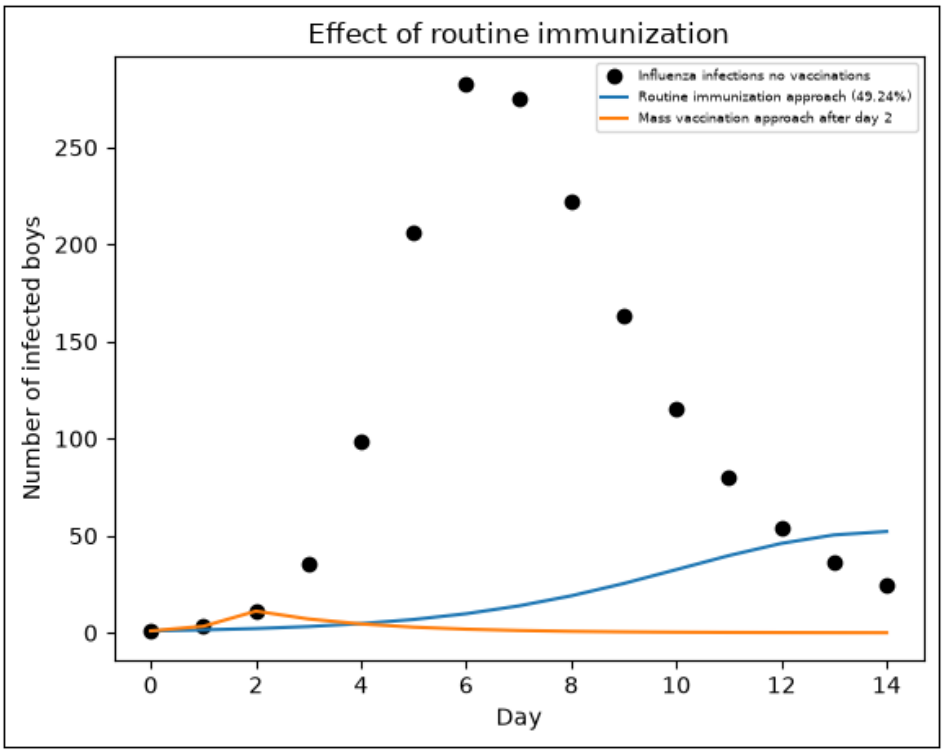


Figure 5. Result of routine immunization and mass vaccination approach compared to no vaccination.

4.4 SIR with demography and equilibria

Adding births and natural deaths changed the dynamics of the SIR model. Figure 6 shows a SIR model with demography and no endemic. For $\beta = 0.2$, $\gamma = 0.3$ and $\mu = 0.1$, $R_0 = 0.50$. The infected population decreased towards zero and the system approached the disease free equilibrium.

Figure 7 shows a SIR model with demography and endemic equilibrium, since $R_0 > 0$ (Keeling & Rohani, 2008, Chapter 2). With $\beta = 1.5$, $\gamma = 0.3$ and $\mu = 0.01$, $R_0 = \frac{1.5}{0.3+0.01} = 4.84$. The infected fraction showed initial oscillations that dampened over time until reaching the equilibrium, in which $I^* = \frac{0.01}{1.5}(4.84 - 1) = 0.256$. With $\beta = 0.49$, $\gamma = 0.3$and $\mu = 0.01$, $R_0 = \frac{0.49}{0.3+0.01} = 1.6$ the system also approached an endemic equilibrium with $I^* = \frac{0.01}{0.49}(1.6 - 1) = 0.012$, but the oscillations persisted for a longer period (Keeling & Rohani, 2008, Chapter 2). This is visualised in Figure 8.

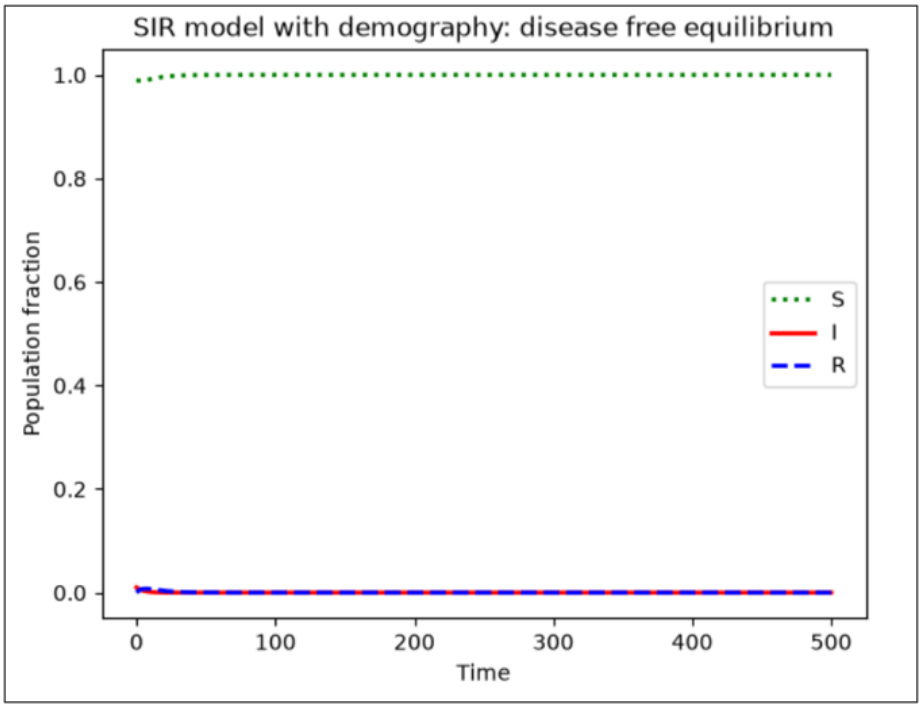


Figure 6. SIR model with demography. Shows disease free equilibrium

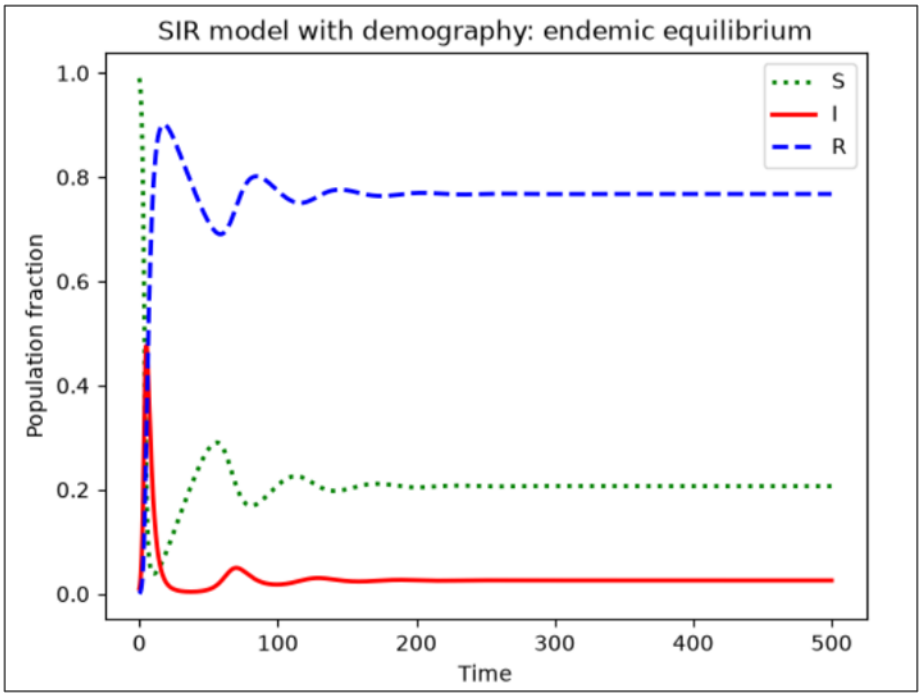


Figure 7. SIR model with demography. Shows endemic equilibrium.

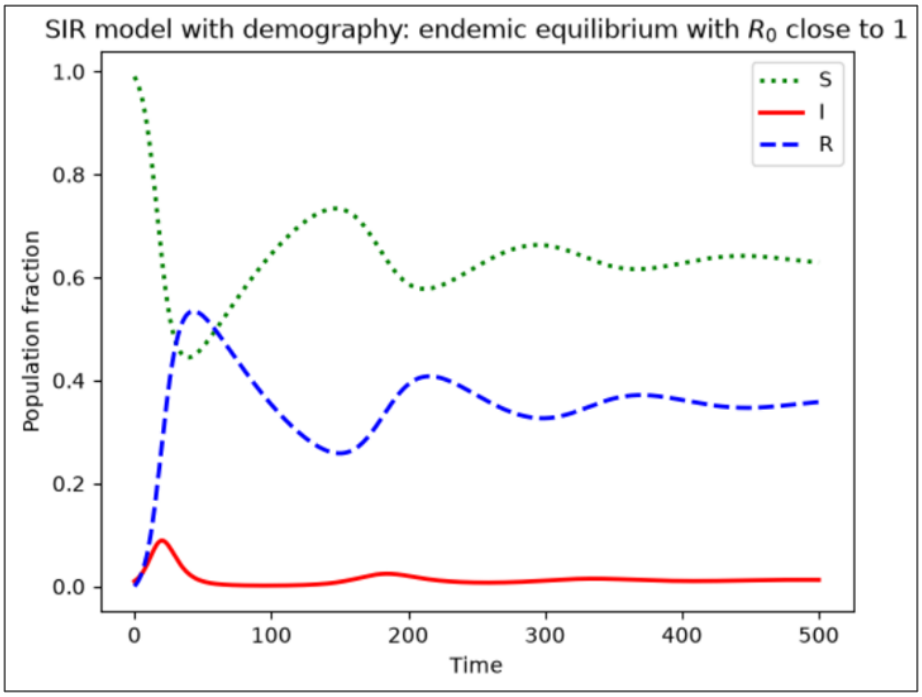


Figure 8. SIR model with demography. R_0 is close to 1, which leads to a longer oscillation period before approaching equilibrium.

Fourier analysis showed one high amplitude peak for the graph that has a $R_0 = 4.84$, while showing multiple peaks with the $R_0 = 1.6$. This is consistent with the longer oscillation period for a R_0 close to 1 (Keeling & Rohani, 2008, Chapter 2). Fourier analysis plots are visualised in Figure 9.

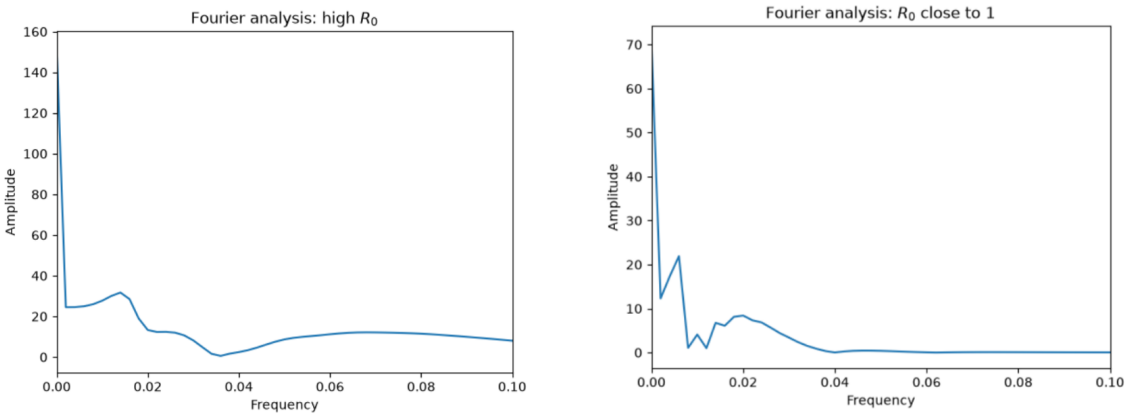


Figure 9. Fourier analysis plots at different R_0 values.

4.5 Infection induced mortality

The effect of infection induced mortality was researched at mortality probabilities of $\rho = 0.1$, 0.5 and 0.99. Increasing ρ affected the epidemic and the total population size. This is displayed in Figure 10. As ρ approaches one, m becomes large. Infected individuals therefore leave the infected compartment quickly (Keeling & Rohani, 2008, Chapter 2). At lower ρ disease persists longer and population size lowers.

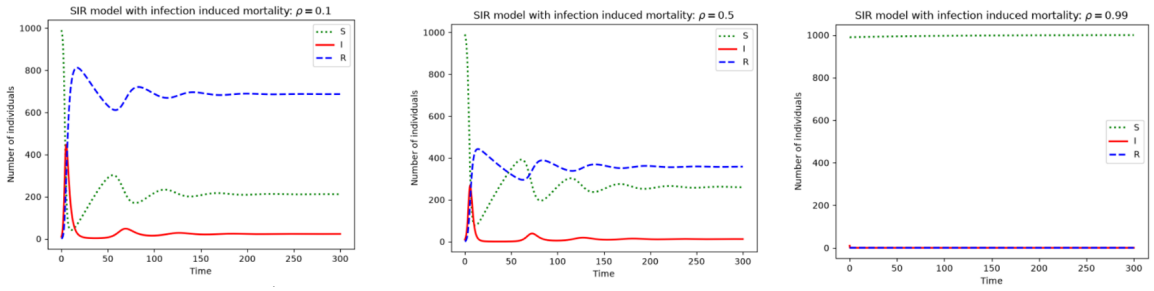


Figure 10. SIR model with demography and infection induced mortality.

4.6 SICR model

The SICR model is visualised in Figure 11. Here $\beta = 1.0,\ \gamma = 0.3,\ k_2 = 0.1,\ k_3 = 0.5,\ \mu = 0.01$. The infected population initially increased and then decreased towards a lower endemic level, while the chronic population increased more slowly and remained stable.

Changing k_2, had an effect on the chronic population (Figure 12) . Larger $k_2$transfers more individuals from the infected compartment into the chronic compartment.

Changing k_3 affects the infected population (Figure 13). With higher k_3 leading to an earlier epidemic peak.

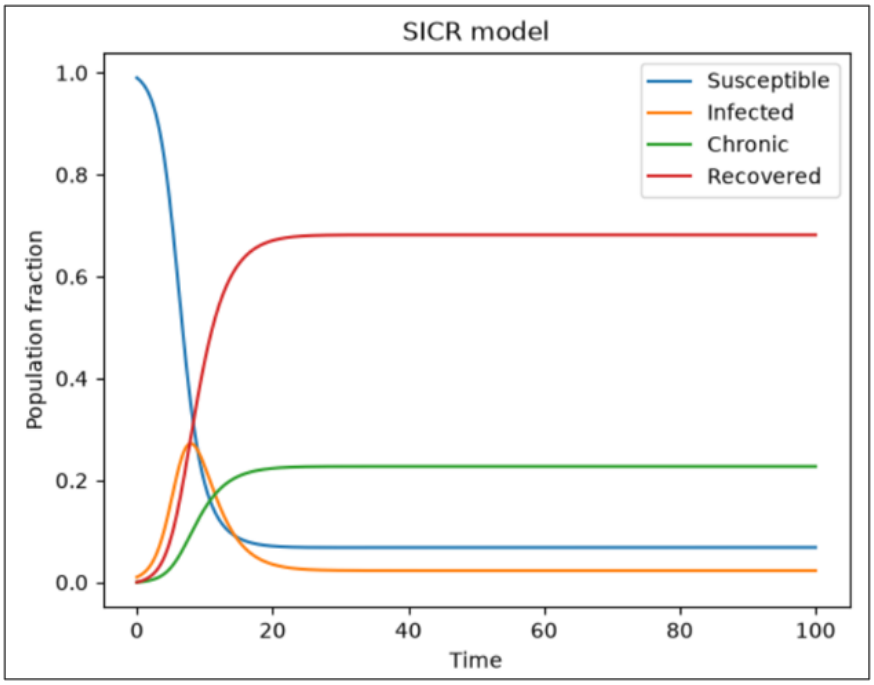


Figure 11. The SICR model with $\beta = 1.0,\ \gamma = 0.3,\ k_2 = 0.1,\ k_3 = 0.5,\ \mu = 0.01$.

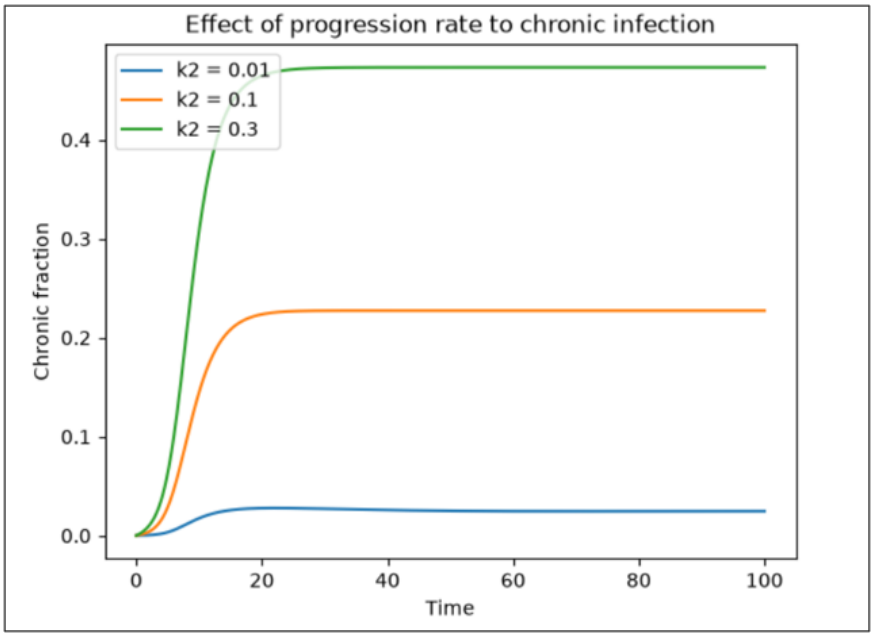


Figure 12. Effect of different progression rates to the chronic fraction in the SICR model

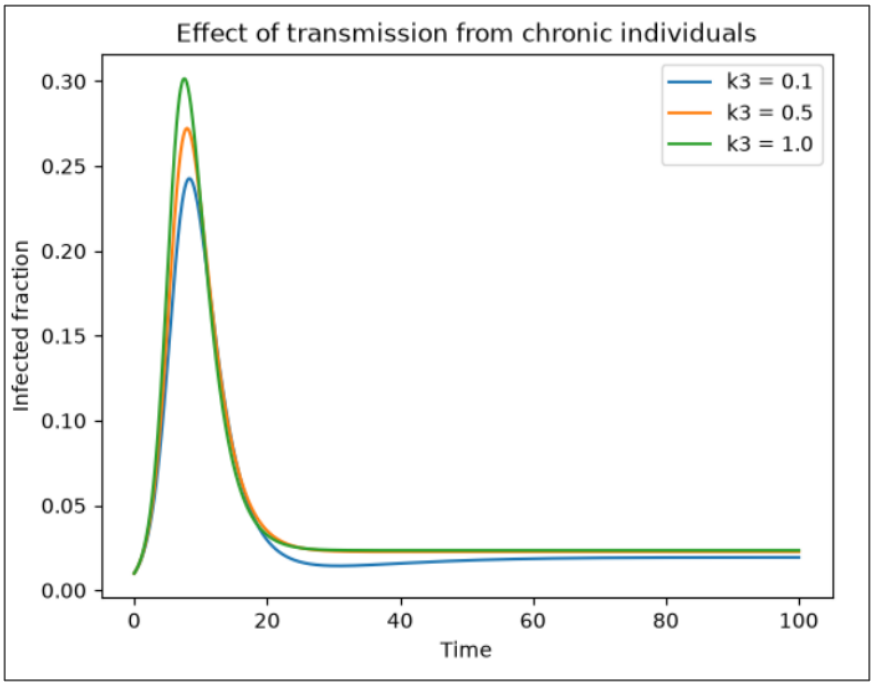


Figure 13. Effect of different transmission rates from chronic individuals on infected fraction in the SICR model.

4.7 Seasonal forcing

The effects of seasonal forcing with different $\beta 1$ on the infected fraction are displayed in figure 14. As $\beta_1$increases, periodic fluctuations appear. The first epidemic peak did not increase continuously with $\beta_1$in these simulations. However, oscillations became stronger as seasonal forcing increased. This is expected because a larger $\beta_1$creates a larger variation in $\beta(t)$ during each 12 month cycle (Keeling & Rohani, 2008, Chapter 5). The results therefore show that seasonal changes can influence amount of infection and timing.

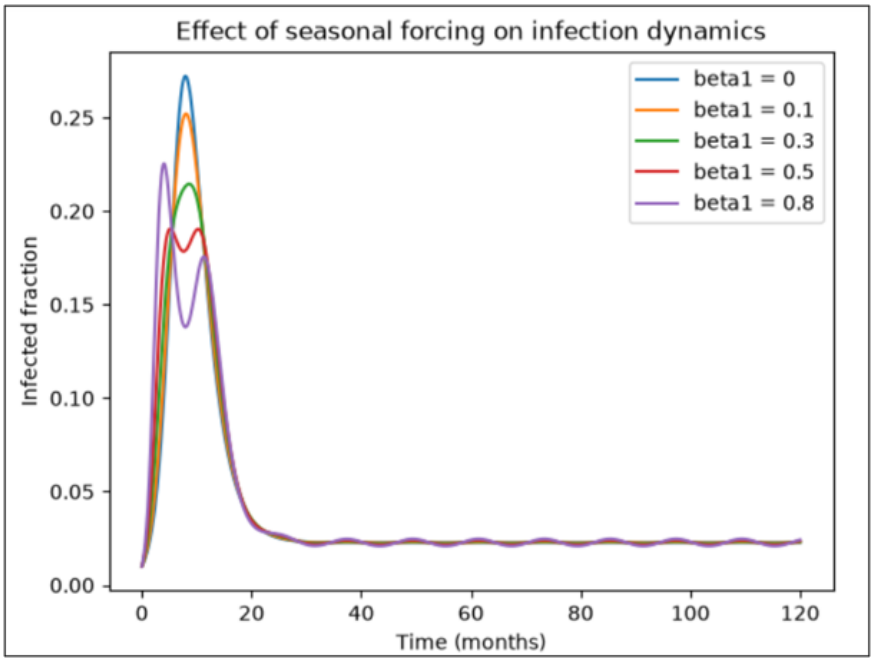


Figure 14. Effect of seasonal forcing on the infected fraction. Different levels of seasonal forcing have been plotted.

4.8 Discussion and future research

This report investigated how changes in epidemiological parameters, control strategies, population dynamics and seasonal transmission affect the behaviour of SIR-based epidemic models. Changes to the SIR models lead to changed dynamics. The SIR model with demography showed oscillations, while the basic SIR model did not. Vaccination strategies lowered the number of infected. The SICR model has an extra compartment, which can also infect susceptible, this also leads to changed dynamics. Seasonal effects had an periodic oscillatory effect on disease transmission.

Even though the fitted basic SIR model described the influenza outbreak in the English boys school relatively well, this model would not work for bigger populations and longer lasting outbreaks, as the basic SIR model assumes homogeneous mixing and does not consider mortality and seasonal effects (Keeling & Rohani, 2008, Chapter 2). While the SIR model with demography includes birth and mortality and in the SICR model seasonal effects were implemented, parameter fitting for a specific disease has not been done. Gnanvi et al. (2021) showed in previous research that COVID-19 models predicted incorrect dynamics. This leads to the conclusion that model assumptions and parameter fitting largely influence dynamics. Future research should focus on model development and parameter fitting for complex diseases to increase predictive accuracy.

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